

Course No. – AS 5570

Principles of Guidance for Autonomous Vehicles

Assignment 1

Due Date: August 31, 2025

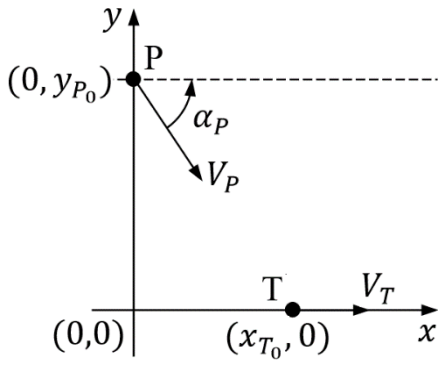
(For Computer Assignments: Sept 7, 2025)

Part A : Introductory Topics

- 1) Distinguish between different levels of autonomy.
- 2) Distinguish between domain-dependent and domain-independent components of autonomous systems.
- 3) Discuss basic differences between path planning, navigation, guidance and vehicle control.
- 4) Draw schematic block diagram of Lateral autopilot and describe its function.
- 5) Draw schematic block diagram of Guidance, Navigation and Control (GNC) system and describe its function.
- 6) Describe different phases of guidance in an engagement.
- 7) Describe non-homing, homing and external guidance implementation schemes.
- 8) What are the different categories of homing guidance in terms of seekers' modes of operation? Give examples.

Part B : No-Maneuver Engagement

- 1) What is a collision triangle geometry? Prove that the closing speed of two unguided vehicles is constant when they are in a collision triangle geometry. Also, show that the time-to-go at any time t can be given as, $t_{go}(t) = -R(t)/\dot{R}(t)$, where $R(t)$ is the range between two vehicles in collision triangle geometry, and $\dot{R}(t) = dR(t)/dt$.
- 2) Consider the engagement between a pursuer (P) and a target (T) with speeds V_P and V_T , respectively. Both the vehicles are non-maneuvering.



- i. Obtain expressions for pursuer's and target's positions $(x_P(t), y_P(t))$, $(x_T(t), y_T(t))$.
- ii. Derive the expression of range between them $R(t)$.
- iii. Derive the conditions on α_P for given $v = V_P/V_T$, x_{T_0} and y_{P_0} such that the range between them decreases for some finite time.
- iv. Under the derived conditions above, obtain the expressions of the miss distance R_{miss} and the time t_{miss} , at which $R = R_{miss}$.
- v. For what value of α_P is R_{miss} zero? What is the time required to achieve zero miss in that case?
- vi. Plot the variation of R_{miss} with $\alpha_P \in [-\frac{\pi}{2}, 0]$ for a given $v = V_P/V_T$. **(Computer assignment)**

3) Consider the trajectories of two vehicles P and T be parameterized with respect to (w.r.t.) time as,

$$X_P(t) = (R_P \cos(\gamma_P t), R_P \sin(\gamma_P t)); \quad X_T(t) = (R_P + R_T \cos(\gamma_T t), R_T \sin(\gamma_T t)),$$

where, γ_P and γ_T are the rate of change of heading angles of P and T. And, $R_P = V_P/\gamma_P$, while $R_T = V_T/\gamma_T$.

- i. Let $\gamma_P = \gamma_T = \gamma$. Justify whether P and T can intercept each other. If 'Yes', what is the final time t_f ? Else, if 'No', what is the miss distance R_{miss} and at which time(s) $R = R_{miss}$?
- ii. Let $\gamma_P \neq \gamma_T$, and $\frac{\gamma_P}{\gamma_T} > 4v$, where $v = V_P/V_T$. Justify whether P and T can intercept each other. If 'Yes', what is the final time t_f ? Else, if 'No', what is the miss distance R_{miss} and at which time(s) $R = R_{miss}$? (If not possible by hand, do the second half of the problem as computer assignment)

Part C : Pure Pursuit (PP) against Non-maneuvering and maneuvering Targets

1) For Pure Pursuit (PP) guidance,

- i. Express range R in terms of angle $\psi \triangleq \alpha_T - \theta$, where α_T is the heading angle of target and θ is the line-of-sight (LOS) angle.
- ii. Show that the trajectory of a PP-guided pursuer in an engagement against a non-stationary non-maneuvering target remains in one half-plane.

- iii. For speed ratio $\nu = \frac{V_P}{V_T} = 1$, show that $R(\psi)$ is parabolic in nature. What is the final distance between the pursuer and target in that case?
 - iv. Express pursuer's lateral acceleration a_p as a function of $\psi \triangleq \alpha_T - \theta$ for $\nu > 1$.
 - v. For $1 < \nu \leq 2$, for which value of ψ is the pursuer's lateral acceleration maximum? And, what is the value of $a_{p_{max}}$?
 - vi. Plot the curves showing variation of R with ψ for $R_0 = 5000\text{m}$ and $\psi_0 = \pi/3$ and $\alpha_{P_0} = \theta_0$ for speed ratios $\nu = 0.8, 1$, and 1.2 . (Computer assignment)
- 2) NPTEL lecture series Question 3 on Page 108-109 for speed ratios $\nu = \frac{V_P}{V_T} = 0.8, 1$ and 1.5 . (It's mainly **Computer assignment**. However, you should practice computations as much as possible in hand.)
- 3) Consider a Pure Pursuit (PP)-guided pursuer with speed V_P against a maneuvering target with speed V_T and constant turn rate $\dot{\alpha}_T$. Analyze the PP trajectory on (V_θ, V_r) -space and find the conditions of successful capture. Identify the capture region in terms of V_{θ_0}, V_{r_0} .
- 4) NPTEL lecture series Question 3 on Page 108-109 for speed ratios $\nu = \frac{V_P}{V_T} = 0.8, 1$ and 1.5 and constant target maneuver $\dot{\alpha}_T = \pm\pi/6$ rad/sec. (It's mainly **Computer assignment**. However, you should practice computations as much as possible in hand.)
- 5) Consider a Pure Pursuit (PP)-guided pursuer with speed V_P against a maneuvering target with speed V_T and turn rate $\dot{\alpha}_T = c_T \dot{\theta}$. Consider three cases of $\nu = V_P/V_T \lesseqgtr 1$. For each of these three cases consider $c_T \lesseqgtr 1$. For each of these 9 cases:
- i. Obtain the trajectory on the (V_R, V_θ) -space.
 - ii. In cases of successful capture:
 - i. What are the conditions of successful capture?
 - ii. Identify the capture region in terms of V_{R_0} and V_{θ_0} .
 - iii. Derive the expression of t_f .
 - iii. In other cases (no capture):
 - i. Derive the expressions of t_{miss} and R_{miss} .
 - iv. Derive expressions of $R(\psi)$ and $a_p(\psi)$. Obtain final values of R and a_p for $\nu \geq 1$.
 - v. Obtain the value of $a_{p_{max}}$, and find at which value of ψ , $a_p(\psi) = a_{p_{max}}$.
- 6) Consider the initial engagement geometry in Question 3 on Page 108-109 of NPTEL lecture series. Consider speed ratios $\nu = 0.8, 1, 1.5$. For each of these three cases consider $c_T = 0.8, 1, 1.5$, where $\dot{\alpha}_T = c_T \dot{\theta}$. Solve Question 3 with all these set-ups.

(Part A: A1, A2, A5 – **Computer assignment**; Part B: C1 – **Computer assignment**)